



A Course Project Report
on
Sentiment Analysis of Airline Tweets

submitted in partial fulfillment of the requirements
for the course

Machine Learning

Submitted by

Name: Sakshya Sinha

Roll No: 2305326

Course / Branch: B.Tech Computer Science and Engineering

Semester: 6th

Submitted to

Course Faculty: Dr. Partha Pratim Sarangi

Department of Computer Science and Engineering

Kalinga Institute of Industrial Technology

Academic Year: 2025–2026



CERTIFICATE

This is to certify that the course project entitled “**Sentiment Analysis of Airline Tweets**” is a bona fide work carried out by **Sakshya Sinha** (Roll No: 2305326), a student of B.Tech Computer Science and Engineering, 6th Semester, Kalinga Institute of Industrial Technology, in partial fulfillment of the requirements for the course **Machine Learning** during the academic year 2025–2026.

The work presented in this project was carried out under my guidance and supervision.

Course Faculty

Dr. Partha Pratim Sarangi

Department of Computer Science and Engineering

Date:

Place: Bhubaneswar

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my course faculty, Dr. Partha Pratim Sarangi, Department of Computer Science and Engineering, Kalinga Institute of Industrial Technology, for his valuable guidance, continuous support, and encouragement throughout the completion of this project.

I am thankful to the faculty members of the Department of Computer Science and Engineering for providing the necessary resources and a conducive learning environment. I would also like to thank my friends and classmates for their cooperation and helpful discussions.

Finally, I express my heartfelt gratitude to my parents for their constant motivation and support, which played a crucial role in the successful completion of this project.

Contents

Certificate	1
Acknowledgement	2
1 Introduction	4
2 Methodology	5
2.1 Dataset Description	5
2.2 Text Preprocessing	5
2.3 Feature Extraction	5
2.4 Classification Model	5
2.5 System Workflow	6
3 Experiments and Results	6
4 Conclusion	6

Sentiment Analysis of Airline Tweets

Student Name: Sakshya Sinha

Roll Number: 2305326

Abstract

Sentiment analysis is an important application of natural language processing and machine learning that focuses on identifying and categorizing opinions expressed in textual data. This project analyzes airline-related tweets to classify customer sentiment into positive, neutral, and negative categories. The primary objective is to gain practical experience in Python programming and foundational machine learning concepts for text classification. The Twitter US Airline Sentiment dataset is used, and the raw text data is processed through standard NLP techniques such as text cleaning, stopword removal, and stemming. Feature extraction is performed using the Bag of Words model, followed by training a Multinomial Naive Bayes classifier. Model performance is evaluated using accuracy, confusion matrix, and classification metrics. The results demonstrate that traditional machine learning models can effectively classify sentiment in social media data while highlighting the limitations of simple text representations.

1 Introduction

The rapid growth of social media platforms has led to an enormous volume of user-generated textual data. Organizations increasingly rely on automated techniques to analyze public opinion and customer feedback. Sentiment analysis, a key application of machine learning, aims to determine the emotional tone of text and has become especially relevant for industries that receive continuous customer feedback.

In the aviation industry, passengers frequently share their experiences on platforms such as Twitter, including feedback related to delays, cancellations, customer service, and overall satisfaction. Manually analyzing such large-scale unstructured data is impractical and time-consuming. Automated sentiment classification provides a scalable and objective solution to this challenge.

Previous research in sentiment analysis has explored lexicon-based methods as well as supervised machine learning approaches such as Naive Bayes, Support Vector Machines, and Logistic Regression. While deep learning models have shown improved performance, traditional machine learning techniques remain effective due to their simplicity and interpretability. This project focuses on developing a supervised machine learning model using Multinomial Naive Bayes to classify airline tweets into positive, neutral, and negative sentiments, while providing hands-on exposure to the complete machine learning workflow.

2 Methodology

This project follows a structured machine learning pipeline consisting of data collection, preprocessing, feature extraction, model training, and evaluation.

2.1 Dataset Description

The Twitter US Airline Sentiment dataset is used in this study. It contains thousands of tweets related to major U.S. airlines, each labeled with a sentiment category: positive, neutral, or negative. The dataset includes raw tweet text along with corresponding sentiment labels.

2.2 Text Preprocessing

Text preprocessing is essential to reduce noise and improve model performance. The following steps were applied:

- Removal of non-alphabetic characters using regular expressions
- Conversion of text to lowercase
- Tokenization of text into individual words
- Removal of stopwords while retaining negation terms such as “not”
- Application of Porter stemming to reduce words to their root forms

2.3 Feature Extraction

The cleaned text data is transformed into numerical form using the Bag of Words (BoW) model. A CountVectorizer is used to extract the top 1500 most frequent words, representing each tweet as a vector of word frequencies.

2.4 Classification Model

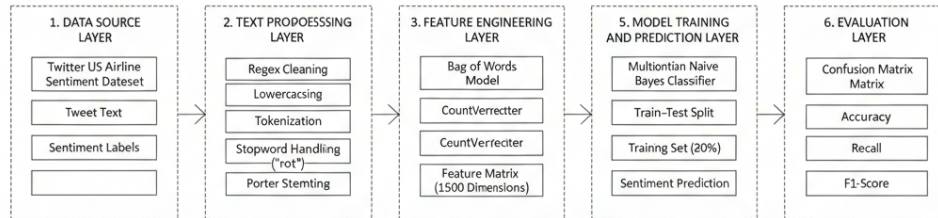
A Multinomial Naive Bayes classifier is employed due to its effectiveness in text classification tasks. The model is based on Bayes’ Theorem:

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

where C represents the sentiment class and X denotes the feature vector. The dataset is split into training and testing sets using an 80:20 ratio.

2.5 System Workflow

The overall system follows a linear workflow starting from raw tweet input, followed by preprocessing, feature extraction, model training, and evaluation. This architecture allows modular development and future enhancements.



3 Experiments and Results

After training the Multinomial Naive Bayes classifier, predictions were made on the test dataset. The model achieved an overall accuracy of 0.60, demonstrating reasonable performance given the simplicity of the approach and the inherent ambiguity in natural language data.

The confusion matrix obtained from the evaluation is shown below:

Actual \ Predicted	Negative	Neutral	Positive
Negative	82	9	10
Neutral	23	16	11
Positive	17	10	22

The classification report is summarized in Table below:

Class	Precision	Recall	F1-score
Negative	0.67	0.81	0.74
Neutral	0.46	0.32	0.38
Positive	0.51	0.45	0.48

The results indicate that the classifier performs best in identifying negative sentiment, which is the dominant class in the dataset. Neutral sentiment remains the most challenging category due to linguistic overlap with both positive and negative expressions. Despite these limitations, the results confirm that traditional machine learning methods combined with effective preprocessing can yield meaningful sentiment classification outcomes.

4 Conclusion

This project demonstrates the application of machine learning techniques for sentiment analysis of airline-related tweets. By implementing a complete supervised learning pipeline, the study provides practical experience with data preprocessing, feature extraction, model training, and evaluation using Python.

The Multinomial Naive Bayes classifier, combined with the Bag of Words model, proves to be a simple yet effective approach for text classification tasks. However, the model is limited in capturing contextual information, word order, and sarcasm, which affects performance on neutral sentiment detection. Future improvements may include the use of TF-IDF features, n-gram models, class balancing techniques, and advanced deep learning architectures to enhance classification accuracy. Overall, the project strengthens foundational understanding of machine learning concepts and prepares students for real-world applications and internship opportunities.